LESSON PLAN

AVIONICS AND ELECTRONIC WARFARE (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	25	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	A/C power supply systems-Constant frequency systems, operation of CSDU, integrated drive generators, field excitation of frequency wild gene
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	A/C power supply systems-Constant frequency systems, operation of CSDU, integrated drive generators, field excitation of frequency wild generators.
5. Specific Objectives	13.1.1.1 Trainee will be able to explain the A/C power systems.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of CSDU and IDG schematic diagrams; Safety: Ensure no live electrical connections during diagram analysis; Instructor supervision: Monitor trainee identification of components.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the aircraft AC power supply system?
(b) List two types of AC power generation systems used in modern aircraft.
(c) Why is constant frequency power required for certain avionics?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Overview of Constant Frequency Systems • Operation of Constant Speed Drive Unit • Integrated Drive Generator Components • Field Excitation of Frequency Wild Generators	L & Ds	PPT, IB	15 min
(b)	• Mechanical coupling of CSD to Engine • Hydraulic control of CSD speed • Generator output regulation mechanisms • Excitation system voltage control	L & Ds	PPT, PDF	20 min
(c)	• Failure modes of CSDU and IDG • Protection systems for AC power • Comparison of Constant vs Variable frequency • Maintenance requirements for generators	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the CSDU maintains constant frequency despite engine speed variations.
(b) Describe the role of field excitation in regulating generator output voltage.
(c) Identify three key differences between constant frequency and variable frequency systems.

12. Home Assignment and Follow Up	Prepare a brief note on the schematic of an external power supply and over voltage protection, linking concepts to the AC systems discussed in this lesson for the next session on DC systems.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Introduction to DC systems: - power supply systems, three pin receptacle systems, schematic of an external power supply and over voltage protection. (Multiple DC bus bar systems). Practical on 3-Pin Receptacle system.	Refer to the master syllabus reference list	LP 06

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min